# One-monomino stabilization in near-square tatami rectangles

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**Abstract**

We extract the fixed-gap one-monomino stabilization law from the generating function of Alhazov, Morita and Iwamoto. The eventual count is four at gap two and zero at larger even gaps. We identify the last exception and distinguish a parity-admissible anchor from an integer cutoff.

## 1 Statement

Let $T(r, w, m)$ be the number of tilings of a labelled $r$-by-$w$ grid by dominoes and exactly $m$ unit squares (monominoes), with no four tile corners meeting at a grid vertex. Reflected or rotated tilings are counted separately whenever they are distinct. For $m = 1$, the area must be odd, so a positive odd gap $d = w - r$ allows no tiling, and for even $d$ only odd $r$ is admissible.

**Theorem 1.** *For $d = 2$, $T(5, 7, 1) = 8$ and $T(r, r + 2, 1) = 4$ for every odd $r \geq 7$, with one tiling for each choice of the monomino's corner. For every even $d \geq 4$, $T(d + 3, 2d + 3, 1) = 4$, and $T(r, r + d, 1) = 0$ for every odd $r \geq d + 5$.*

For even $d \geq 2$, the least admissible stabilization anchor is $d + 5$. The least integer cutoff beyond which the admissible subsequence has its eventual value is $d + 4$. An anchor must itself have the prescribed parity.

## 2 Literature input

For odd $r \geq 5$ and $w \geq r$, use Alhazov, Morita and Iwamoto's one-monomino enumeration [1, §§3, 5 and 7]. In the variable $z$ marking width, put $P = z^{r-1} + z^{r+1}$. The count at width $w$ is the coefficient of $z^w$ in $B + C$, where

$$B = \frac{4(z^{r-2} + z^r)}{1 - P}, \qquad C = \frac{2(z + z^r)}{(1 - P)^2}.$$

In $B$, the numerator marks the two special corner block widths. In $C$, the numerator marks the single-column block $L_r$, which has a monomino at one end, and the width-$r$ block $M_r$; the two factors $(1 - P)^{-1}$ describe lists of pure domino blocks on either side. We use the widths and formulas from §7: the second shift in the printed §6 recurrence for $B$ and the inequality introducing §7 are inconsistent with the block enumeration and intended longer-width range.

## 3 Extracting the fixed-gap coefficients

Expand $(1-P)^{-1}$ as $\sum_{k\geq 0} P^k$ and $(1-P)^{-2}$ as $\sum_{k\geq 0}(k+1)P^k$. A single $P$ contributes exponent $r-1$ or $r+1$; two or more factors have exponent at least $2(r-1)$. At $w = r+d$, the coefficient offsets from the two monomials in $B$ are $d+2, d$; the offsets in $C$ are $r+d-1, d$.

For $d = 2$ and odd $r \geq 7$, both offsets in $B$ and the second offset in $C$ are positive and less than $r-1$. The first offset in $C$ is $r+1$, which comes from one $P$ factor and yields $2 \cdot 2 = 4$. These tilings use $L_r$ and one width-$(r+1)$ block, so their monominoes are at the corners. The two axis reflections act transitively on corners, giving one tiling at each. At $r = 5$ the same $C$ term contributes four, and $B$ contributes another four from $d+2 = r-1$.

For even $d \geq 4$ and odd $r \geq d+5$, both $B$ offsets and the second $C$ offset are smaller than $r-1$ and positive. The first $C$ offset satisfies $r+1 < r+d-1 < 2(r-1)$, so it contributes nothing either. At $r = d+3$, the first $B$ offset is $r-1$, contributing four. The other $B$ offset and second $C$ offset are below $r-1$; the first $C$ offset is $2r-4$, strictly between $r+1$ and $2(r-1)$ since $r \geq 7$. This proves both the eventual value and the last exception.

## 4 What this says about the conjecture

The last exceptional admissible size is $d+3$ for every positive even $d$, so the next admissible size $d+5$ is the least anchor. The cutoff need only exclude $d+3$, giving $d+4$. For instance, $T(7, 11, 1) = 4$, and all counts from $r = 8$ onward are zero at $d = 4$. Even if one anchors to the value at the cutoff itself, the literal integer minimum here is 8, whereas the experimental expression $m+d+4$ in the formulation of Erickson et al. [2, §4.1, Conjecture 3] gives 9. Restricting the anchor to admissible sizes gives 9 and removes this discrepancy. With $d$ odd there is no admissible subsequence for $m = 1$. The square case is covered separately by Erickson and Schurch [3].

This note uses a published enumeration and does not resolve the general fixed-$m$ conjecture. The companion script checks finite coefficient examples.

**AI contribution.** GLM5.3F (source project credit supplied by the author). Literature verification and preparation used GPT Sol Max assistance.

## References

[1] A. Alhazov, K. Morita and C. Iwamoto, A note on tatami tilings, *RIMS Kôkyûroku* **1691** (2010), 1–7. Kyoto University archive.

[2] A. Erickson, F. Ruskey, M. Schurch and J. Woodcock, Auspicious tatami mat arrangements, arXiv:1103.3309 (2011).

[3] A. Erickson and M. Schurch, Monomer-dimer tatami tilings of square regions, *J. Discrete Algorithms* **16** (2012), 258–269. arXiv:1110.5103.